

[SECTION-A BIOLOGY]**MM:09**

Q.1 If the plasma membrane of a cell is completely damaged, the cell will most likely: 1

- a) Continue normal functioning.
- b) Lose control over the movement of substances and eventually die.
- c) Produce more chlorophyll.
- d) Divide rapidly.

- Q.2
- A. Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A.
 - B. Both Assertion (A) and Reason (R) are true, but R is not the correct explanation of A.
 - C. Assertion (A) is true, but Reason (R) is false.
 - D. Assertion (A) is false, but Reason (R) is true.

Assertion (A): Chromosomes are found inside the nucleus.

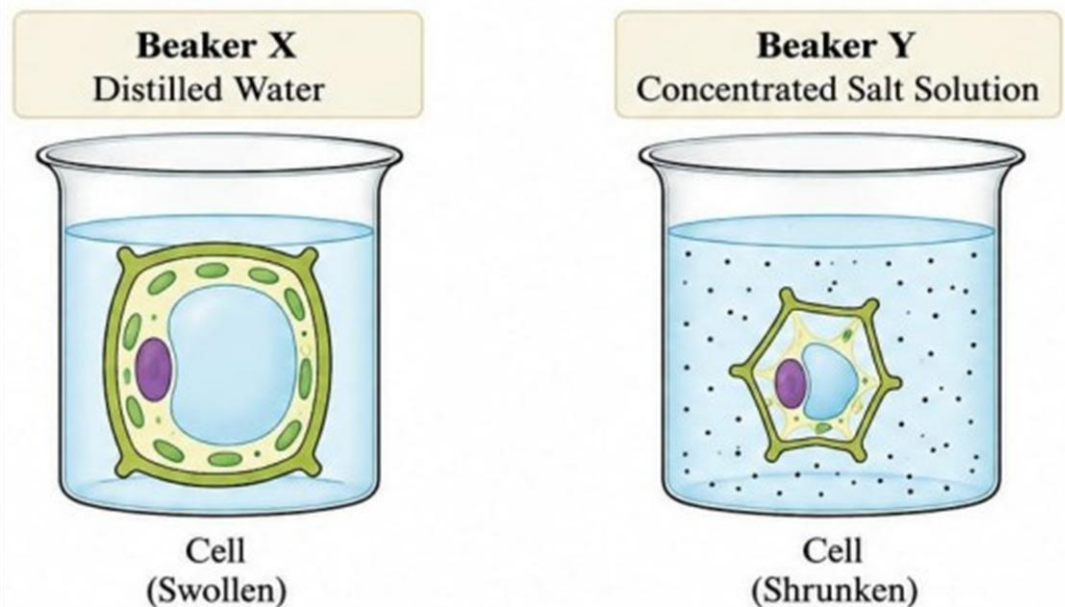
Reason (R): Chromosomes contain DNA and genes that carry hereditary information.

1

Q.3 Case-Based Question:

In a biology laboratory, students were given an activity to study the effect of different solutions on plant cells. The teacher placed two similar plant cells in two different beakers containing different solutions and left them for some time. The students observed that in Beaker X, the cell became swollen, while in Beaker Y, the cell became shrunken.

The figure below shows the observations made by the students.



Based on the above information, answer the following questions.

- a) Which beaker contains a hypotonic solution? 1
 - b) Which beaker contains a hypertonic solution? 1
 - c) Name the process responsible for the change in both cells. 1
 - d) Explain why the cell in Beaker Y became shrunken. 1
- Q.4 Draw a neatly labelled diagram of a plant cell and label any 3 parts. 3

[SECTION-B PHYSICS]**MM:08**

Q1. Choose the correct answer

A boy walks 6m north then 8m east. What is the magnitude of his displacement? 1

- (a) 14 m (b) 10 m (c) 2 m (d) 48 m

Q2. Assertion (A): Average velocity of an object can be zero even if Average speed is not Zero.

Reason (R): Average velocity depends on displacement while average speed depends on total distance travelled. 1

Q3. Write True/False:

An object in uniform circular motion has zero acceleration. 1

Q4. A car starts from rest and accelerates Uniformly to 20 m/s in 5 sec. It then travels at 20 m/s for 10 sec and finally applies the breaks (with uniform acceleration)to stop in 6 sec. Find the total distance travelled. 2

Q5. Derive the kinematic equation for Uniformly accelerated motion from a velocity-Time graph.

$$V^2 = U^2 + 2as$$
 3

[SECTION-C CHEMISTRY]**MM:08**

Q6. Which method is used to separate cream from milk? 1

- (a) Filtration (b) Evaporation (c) Centrifugation (d) Sublimation

Q7. A student prepared a Glucose solution by dissolving 20 g of Glucose in 180 g of water. Calculate the mass percentage of Glucose in the solution. 2

Q8. State any two differences between a True Solution & a Colloid. 2

Q9. i) Define Coagulation. 1+2

ii) How would you separate a mixture of common salt & ammonium Chloride? Explain the principle involved.

